

FRE-GY 9743 Assignment 1

Bond Forward Business

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I use prices per dollar of face value and write

$$N = 1,000,000, \quad S_t = B(t; t, T_m), \quad F_t = B(t; T_s, T_m),$$

so S_t is the dirty spot price and F_t the dirty forward price for delivery at T_s . Coupons per dollar of face are C_i , paid at t_i . For the repo leg I write the simple accumulation factor

$$A_{R,t}(a, b) = 1 + R_t \tau(a, b),$$

with τ on Act/360 and R_t the term repo rate observed at t . The CSA discount factor is $P^c(t, T_s)$.

Q1

Question. *What could motivate the client to buy this forward? What are the important contract terms, and what happens at settlement?*

Answer.

The client is fixing today the price it will pay a year from now. The typical case is an asset manager that knows cash arrives at T_s and will be invested in this Treasury: if yields fall in the meantime the bond gets more expensive, and the forward removes that risk. Two other motives are worth naming. First, the exposure is *unfunded*, since the client pays nothing at inception beyond variation margin, so it gets a year of duration without paying S_0 today and without using its own repo lines or balance sheet. Second, since K is just spot plus carry, buying the forward is also a view that the implied financing is too high, e.g. that the bond will go special. It remains a commitment, not an option: if yields rise, the client still buys at K .

The contract should state the exact Treasury (CUSIP, coupon, maturity), the \$1,000,000 face notional, trade date and T_s , the strike K and whether it is quoted clean or dirty, physical or cash settlement, coupon entitlement around T_s , business-day and day-count conventions, and the CSA terms. “A 10-year Treasury” is not specific enough, because different issues have different coupons and prices, and I do not want that ambiguity in my book. The CSA terms matter more than they look: they set P^c and therefore feed directly into the MTM in Q4.

For physical settlement, Bank A delivers N face of the bond and the client pays NK when K is dirty per dollar of face. If quoted clean per 100 face, the invoice is

$$\frac{N}{100} \left(K_{\text{clean},100} + AI_{100}(T_s) \right).$$

For cash settlement the signed payment to the client is $N(B_{\text{ref}}(T_s) - K)$. If that amount is negative, the client pays Bank A.

Q2

Question. *Why does the desk buy the spot bond at inception, and how is that position funded through the treasury and repo desks?*

Answer.

Selling the forward leaves the desk short the bond for delivery at T_s . If the bond rallies I have to deliver something worth more than the fixed strike, so unhedged I am carrying the full duration and convexity of a 10-year Treasury on \$1mm of face for a year, which is risk I am not paid to take. Buying N face of the exact bond today at NS_0 turns the obligation into a *static replication*: I already hold what I have promised, so there is nothing to rebalance and no path dependence. The only remaining unknown is the cost of carrying the bond to T_s , and Q3 removes that too. After both legs are on, the forward price is pinned down by quantities observable today and I never have to forecast where rates go.

Funding runs through two internal trades. The treasury desk supplies the cash at $t = 0$ to settle the purchase, unsecured and at the internal FTP rate, so it serves only as a bridge, and it also funds whatever the repo haircut leaves uncovered. The rates desk then passes the bond to the repo desk as collateral and borrows against it in a *term* repo to T_s , using that cash to repay the bridge. Making the repo term rather than rolling overnight is the point: the financing cost is then known at inception, whereas a roll would leave me exposed to repo moves and to the bond going special. At T_s I repay the repo balance, take the bond back, deliver it and receive NK . Coupons received before settlement reduce the financing balance; they are carry, not profit.

Q3

Question. *Using the bond's coupon schedule, the term repo rate R , and the spot price, how do we compute the forward bond price?*

Answer.

It is easiest to treat the repo as a loan balance. I borrow S_0 to buy the bond, and that borrowing accrues to $S_0 A_{R,0}(0, T_s)$ at settlement. When a coupon arrives at t_i it is passed back to me as collateral giver and pays down the loan, which also saves the repo interest that would have accrued on that amount from t_i to T_s . So coupons do reduce the debt, each by $C_i A_{R,0}(t_i, T_s)$. Hence

$$F_0 = S_0 A_{R,0}(0, T_s) - \sum_{0 < t_i < T_s} C_i A_{R,0}(t_i, T_s), \quad (1)$$

and the fair strike is $K = F_0$, so the client pays $NK = NF_0$.

This is the no-arbitrage level: at T_s I own the bond, owe the right-hand side of (1), and receive K . If K were above it I would bank a riskless profit having invested nothing at $t = 0$; if below, the client would. With no coupons it collapses to $S_0(1 + R\tau(0, T_s))$, the borrowing plus repo interest.

Three conventions are worth stating. S_0 is the *dirty* price, since that is the cash I actually borrow, so (1) returns a dirty forward; clean per 100 face is $K_{\text{clean},100} = 100F_0 - AI_{100}(T_s)$. I assume repo funds the full dirty price with no haircut, since only R is given, so any haircut needs separate treasury funding and its cost must be added. And a coupon falling exactly on T_s is included or not depending on the settlement convention, which is why I write the sum strictly.

Writing $P^R(0, t) = 1/A_{R,0}(0, t)$ puts (1) in cash-and-carry form,

$$F_0 = \frac{S_0 - \sum_i C_i P^R(0, t_i)}{P^R(0, T_s)},$$

from which the forward sits below spot whenever the coupon yield exceeds the repo rate, i.e. whenever carry is positive.

Q4

Question. *How do we compute the mark-to-market at $t \in (0, T_s]$?*

Answer.

I redo Q3 with the information available at t : the current dirty price S_t , the repo curve then observed, and only the coupons still unpaid:

$$F_t = S_t A_{R,t}(t, T_s) - \sum_{t < t_i < T_s} C_i A_{R,t}(t_i, T_s).$$

The client is long, so

$$V_{\text{client}}(t) = NP^c(t, T_s)(F_t - K). \quad (2)$$

The strike K never moves, since it was fixed at inception, so all the variation is in F_t . The sign is sensible: if the forward has risen above the locked strike the contract is valuable to the client, and Bank A has the opposite value.

The two curves are doing different jobs and should not be conflated. The repo curve is the *financing* rate that generates F_t , while the CSA curve is the *collateral remuneration* rate that discounts the T_s cash flow back to t . Any spread between them is a genuine risk I run. At settlement $P^c(T_s, T_s) = 1$ and $F_{T_s} = B(T_s; T_s, T_m)$, recovering the payoff.

Q5

Question. *Is the desk left with major market risk, and how can it earn revenue while still offering the client a sensible trade?*

Answer.

With the exact bond owned and repo locked to T_s , every leg matures on the same date and the first-order Treasury risk is hedged. If the bond rallies the short forward loses and the bond gains, so DV01, convexity and curve exposure net to roughly zero. What remains is second-order but not nothing:

- **Counterparty credit (CVA)**, the largest residual. If the client defaults before T_s my hedge becomes a naked long bond to be liquidated at whatever the market then is.
- **Repo basis and specialness**, if the repo is rolled rather than termed, plus the structural basis between the repo curve building F_t and the CSA curve discounting it.
- **Collateral and liquidity risk**, since CSA and repo margin calls are not synchronised, so a matched book can still face funding gaps.

- **Balance sheet and leverage-ratio cost** of carrying \$1mm of Treasuries and the repo for a year, plus settlement and fails risk.

Being hedged does not make the trade free. Let $K_{BE} = F_0(R_{\text{bank}})$ be my break-even strike at my own repo cost. For a client buying, I quote

$$K_{\text{client}} = K_{BE} + m, \quad m > 0,$$

earning m per dollar of face, or Nm gross before other costs, riskless and known today. Since (1) is increasing in R , this is equivalently a forward built off a higher financing rate.

The comparison that matters to the client is not my cost but its own alternative. To replicate, it would buy and fund the bond itself, and most clients fund well above GC, with many having no direct repo access, or facing haircuts and wider bid-offer. So as long as

$$F_0(R_{\text{bank}}) < K_{\text{client}} < F_0(R_{\text{client}}),$$

the client gets a better all-in price than it could arrange alone, needs no cash today, and knows its future purchase price, while I keep a margin over my own funding. The profit comes from better repo access and from warehousing the position, its operational risk and the client's credit risk, not from leaving bond risk unhedged. Quoting it transparently as spot plus carry at a stated rate makes the economics legible and lets the client benchmark it, which is the negotiation I want.

Q5 (Starred)

Question. *How are Q2–Q3 related to risk-neutral pricing?*

Answer.

They are two routes to the same strike. In Q2–Q3 I follow the actual cash: buy the bond, fund it in repo, use coupons to pay down the borrowing. In risk-neutral pricing I take an expectation of the settlement value.

Under perfect collateralisation a newly struck forward has zero value:

$$0 = \mathbb{E}_0^{\mathbb{Q}^c} \left[D^c(0, T_s) N(B(T_s; T_s, T_m) - K) \right],$$

where D^c is the CSA discount factor and N cancels. Changing to the T_s -forward measure,

$$K = \mathbb{E}_0^{\mathbb{Q}^{T_s, c}} [B(T_s; T_s, T_m)],$$

so the strike is the expected settlement value under that measure, which is to say the forward price is a $\mathbb{Q}^{T_s}$ -martingale. Applying the same argument to the traded coupon bond gives

$$K = \frac{S_0 - \sum_{0 < t_i < T_s} C_i P^c(0, t_i)}{P^c(0, T_s)},$$

which is (1) with P^R replaced by P^c . Note that no distributional assumption about rates was needed: static replication pins the expectation down without specifying a model.

That substitution is exactly the catch. The two agree when

- (i) repo and collateral rates coincide and are deterministic, $P^R = P^c$. In practice repo/OIS basis and specialness break this, and the two answers separate by precisely that basis;
- (ii) markets are frictionless, so the bond can be bought, shorted and repo'd in any size with no haircut, bid-offer or balance-sheet charge;
- (iii) there is no credit risk and collateralisation is continuous and perfect, so CSA discounting really is the funding of the payoff;
- (iv) coupons are deterministic and the repo is termed at inception. With stochastic rolled financing the replication is no longer static and a convexity adjustment appears.